

Project Design Phase-II

Functional Requirements

Following are the functional requirements of the proposed solution.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	Data Source & Import Set Configuration	* Load external spreadsheet files (CSV, XLSX) into ServiceNow staging tables. * - Auto-detect file format, delimiter, and header rows during creation. * - Support file attachments and direct server file upload methods.
FR-2	Transform Mapping & Field Mapping	* Map staging columns directly to target table fields using 'Auto Map Matching Fields'. * - Support manual custom field mapping and source-to-target field pairing. * - Define Coalesce fields (e.g., Email, Sys ID, Serial Number) to update existing records or prevent duplicates.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-3	Scripted Transformation & Logic	* Execute OnBefore scripts to validate data and format field types (dates, phone numbers, choices). * - Execute OnAfter scripts to perform post-import updates or record links. * - Support dynamic reference field lookup using source values.
FR-4	Data Quality, Logging & Error Handling	* Capture detailed import logs for inserted, updated, skipped, and error rows. * - Provide roll-back capabilities for failed import set runs. * - Enforce mandatory field checks and reject invalid spreadsheet rows without halting the entire batch.

Non-Functional Requirements

Following are the non-functional requirements of the proposed solution.

FR No.	Non-Functional Requirement	Description
NFR-1	Usability	The data import and mapping interface should be intuitive with automated field matching. Transform Maps should provide clear mapping feedback and detailed execution logs for easy troubleshooting.
NFR-2	Security	Import operations must enforce ServiceNow Access Control Lists (ACLs) and role-based permissions (admin, import_admin). Sensitive data in spreadsheets must be handled securely without plain-text exposure in logs.
NFR-3	Reliability	Transform Maps and scripts must execute consistently across system updates without failing. The system should gracefully handle null/empty values, unexpected special characters, and missing fields.

FR No.	Non-Functional Requirement	Description
NFR-4	Performance	Import processing should be optimized for speed, transforming up to 10,000 staging records in under 2 minutes. Batch processing and index utilization should ensure no system lag during execution.
NFR-5	Availability	The Import Set and Transform Map engine must maintain 99.9% availability on the ServiceNow platform with zero downtime caused by script/mapping execution errors.
NFR-6	Scalability	The solution must support large-scale enterprise data imports (100k+ rows) and be easily reusable for multiple target tables (User, Asset, Incident, CMDB) with minimal configuration changes.